
Joint-CS: Simultaneous Illumination Normalization and Sparse Image Compression via Unfolded Recovery

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Abstract

The traditional camera pipeline samples at Shannon–Nyquist rates, compresses to a sparse representation, and only afterwards performs illumination correction on the decoded image. This sequential ordering discards information: sparse recovery applied to a signal whose dynamic range is dominated by an unknown multiplicative illumination field has weaker recovery guarantees than the same recovery applied to the underlying reflectance. We formulate *Joint-CS*, a single inverse problem in the joint variables of DCT-domain reflectance coefficients and a smooth per-column illumination field, and solve it by block-coordinate alternation between a learned-iterative-shrinkage step on the coefficients and a closed-form ridge step on the illumination. Across four experiments on a 5-image natural-scene set—an empirical Donoho–Tanner phase transition, a multi-image rate-distortion sweep, a deep-unfolded LISTA solver, and a per-scene ablation against a sequential CS-then-illumination-correct baseline—we find that the joint formulation outperforms the sequential one by up to +4.7 dB PSNR on textured scenes (cameraman, astronaut) under an $8\times$ illumination gradient, and that a $K=10$ unrolled LISTA matches what FISTA needs 22 iterations to reach. We also report a clear failure mode: on near-uniform scenes (moon) the joint g -update lacks signal to estimate gain from and the method degrades by ~ 8 dB—a scene-dependence that, to our knowledge, the prior CS+ISP literature does not report.

1 Introduction

Modern imaging pipelines acquire N pixels via Nyquist sampling, then immediately throw most of that information away: a typical JPEG keeps only $\sim 5\text{--}10\%$ of the DCT coefficients above the quantization floor, and the image signal processor (ISP) further reshapes the data through demosaicing, tone-mapping, and white-balance correction. Compressed sensing (CS) reverses this order. If the signal $\mathbf{x}$ is S -sparse in some basis Ψ , then $M \ll N$ random linear measurements $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$ suffice to recover $\mathbf{x}$ exactly (up to noise) by solving an ℓ_1 minimization, provided $\mathbf{A}$ obeys the restricted isometry property [? ?].

CS has been extensively studied for compression. It has been less extensively studied as an integrated stage of a RAW image-signal pipeline. The crux of the issue is that a RAW sensor measures the product

$$\mathbf{x} = \mathbf{s} \odot \mathbf{g}, \tag{1}$$

where $\mathbf{s}$ is the underlying scene reflectance (sparse in the DCT) and $\mathbf{g}$ is a spatially-varying illumination field (smooth, not sparse). A sequential pipeline—CS recovery of $\mathbf{x}$, then a downstream

divide-by-illumination step—wastes the prior that s , not x , is sparse in Ψ . Multiplying a DCT-sparse signal by an arbitrary illumination field destroys the sparsity that the ℓ_1 regularizer is trying to exploit; the recovered x then carries an illumination-shaped bias that no downstream column-mean correction can fully undo.

Contributions. We make three contributions:

1. We formulate *Joint-CS*: a joint ℓ_1 /smooth-prior objective in (c, g) that performs sparse recovery and illumination correction in a single optimization, and we derive a block-coordinate solver with a FISTA inner loop on c and a closed-form ridge step on g .
2. We deep-unfold the c -step into a learned $K=10$ -layer LISTA network [?] and stabilize its training under sustained Adam with gradient clipping and best-checkpoint restoration.
3. We benchmark the full system against OMP, ISTA, FISTA, and ADMM baselines across a phase-transition grid, a rate-distortion sweep on a 5-image natural-scene set, and a per-scene $8\times$ -illumination-gradient ablation. We report scene-dependent results—+1 to +3 dB gains on textured scenes, parity on moderate-content scenes, and a clear failure mode on near-uniform scenes—and analyze the mechanism behind it.

2 Mathematical Preliminaries

2.1 Measurement model

The CS forward model treats image acquisition as a linear measurement:

$$y = Ax + w, \quad y \in \mathbb{R}^M, \quad A \in \mathbb{R}^{M \times N}, \quad x \in \mathbb{R}^N, \quad \|w\|_2 \leq \epsilon, \quad (2)$$

with $M \ll N$. The sensing matrix A models the physical encoder (a coded-aperture mask, a random projection circuit, or in this work a column-normalized i.i.d. Gaussian ensemble), and w collects shot, read, and quantization noise.

2.2 Sparsity prior

The unknown signal is S -sparse in some basis Ψ , i.e. $x = \Psi c$ with

$$c \in \Sigma_S \equiv \{c \in \mathbb{R}^N \mid \|c\|_0 \leq S\}. \quad (3)$$

In this work Ψ is the 2D DCT, a strong sparsifying basis for natural images.

2.3 Convex relaxation

Direct minimization of the ℓ_0 count is combinatorial. Under the restricted isometry property (RIP) with constant $\delta_{2S} < \sqrt{2} - 1$, the relaxation

$$c_1^\epsilon = \arg \min_c \|c\|_1 \quad \text{s.t.} \quad \|A\Psi c - y\|_2 \leq \epsilon \quad (4)$$

recovers c to within $O(\epsilon)$ [?]. Equivalently, the Lasso form

$$c_\lambda = \arg \min_c \frac{1}{2} \|A\Psi c - y\|_2^2 + \lambda \|c\|_1 \quad (5)$$

is solved efficiently with proximal-gradient methods.

2.4 Classical solvers

OMP. Greedy orthogonal matching pursuit [?] iteratively selects the column of $A\Psi$ most correlated with the current residual and re-fits the active support by least squares. OMP can recover supports up to $S \lesssim M/\log N$ and is exact on the support but fails abruptly when the support density grows.

ISTA / FISTA. ISTA is the proximal-gradient method for the Lasso:

$$c^{(k+1)} = \mathcal{S}_{\lambda/L} \left(c^{(k)} - \frac{1}{L} (A\Psi)^\top (A\Psi c^{(k)} - y) \right), \quad (6)$$

where $L = \|(A\Psi)^\top A\Psi\|_2$ and $\mathcal{S}_\tau(z) = \text{sign}(z) \max(|z| - \tau, 0)$ is the soft-threshold operator (the prox of $\tau \|\cdot\|_1$). FISTA [?] augments (6) with a Nesterov momentum term and improves the objective-gap rate from $O(1/k)$ to $O(1/k^2)$.

ADMM. For the same objective, ADMM [?] splits $\mathbf{c} \rightarrow (\mathbf{c}, \mathbf{z})$ with constraint $\mathbf{c} = \mathbf{z}$ and alternates a least-squares update on $\mathbf{c}$ (solved once via a cached Cholesky factorization of $(\mathbf{A}\Psi)^\top \mathbf{A}\Psi + \rho \mathbf{I}$) with a soft-threshold update on $\mathbf{z}$, plus a dual ascent on the multiplier.

3 Joint-CS

3.1 Image-formation model with illumination

We assume the RAW signal follows (??). A conventional pipeline first recovers $\mathbf{x}$ from $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$, then estimates and divides out $\mathbf{g}$. The ordering wastes the prior that $\mathbf{s}$ is sparse in Ψ but $\mathbf{x}$ is not: pointwise multiplication by an illumination field destroys DCT-domain sparsity, the ℓ_1 regularizer over-shrinks the coefficients corresponding to high-illumination columns, and the post-hoc correction can only rescale (not restore) what was lost.

3.2 Joint objective

We instead minimize a single objective in $(\mathbf{c}, \mathbf{g})$:

$$\min_{\mathbf{c}, \mathbf{g}} \frac{1}{2} \|\mathbf{A} \text{diag}(\mathbf{g}_{\text{pix}}) \Psi \mathbf{c} - \mathbf{y}\|_2^2 + \lambda_c \|\mathbf{c}\|_1 + \lambda_g \|\nabla \mathbf{g}\|_2^2, \quad (7)$$

where $\mathbf{g}_{\text{pix}}$ tiles a per-column gain vector $\mathbf{g} \in \mathbb{R}^{\sqrt{N}}$ over rows, and ∇ is a 1D discrete gradient that enforces smoothness on the illumination. The problem is jointly non-convex but block-convex.

c-step. With $\mathbf{g}$ fixed, the effective sensing matrix is

$$\mathbf{A}_{\text{eff}}(\mathbf{g}) = \mathbf{A} \text{diag}(\mathbf{g}_{\text{pix}}) \Psi, \quad (8)$$

and the $\mathbf{c}$ -update is a standard Lasso, solved with FISTA (Sec. ??).

g-step. With $\mathbf{c}$ fixed, let $\hat{\mathbf{s}} = \Psi \mathbf{c}$ and reshape to $\hat{S}_{ij}$. The measurement model becomes linear in $\mathbf{g}$:

$$\mathbf{y} = \mathbf{B}(\hat{\mathbf{s}}) \mathbf{g} + \mathbf{w}, \quad B_{m,j}(\hat{\mathbf{s}}) = \sum_i A_{m,i\sqrt{N}+j} \hat{S}_{ij}, \quad (9)$$

and the $\mathbf{g}$ -update is the closed-form ridge solution

$$\mathbf{g}^* = (\mathbf{B}^\top \mathbf{B} + \lambda_g \mathbf{L})^{-1} \mathbf{B}^\top \mathbf{y}, \quad (10)$$

where $\mathbf{L} = \nabla^\top \nabla$ is the 1D discrete Laplacian. We clamp $\mathbf{g} \geq \epsilon_g$ to avoid divide-by-zero.

Gain ambiguity. The model has an inherent scalar ambiguity $(\mathbf{s}, \mathbf{g}) \rightarrow (a\mathbf{s}, \mathbf{g}/a)$ for any $a > 0$. We resolve it at evaluation time by computing the optimal positive gain $a^* = \langle \hat{\mathbf{s}}, \mathbf{s}_{\text{true}} \rangle / \|\hat{\mathbf{s}}\|^2$ and apply the same rule to the sequential baseline.

3.3 Deep-unfolded inner solver

A practical limitation of ISTA / FISTA is the iteration count—hundreds of iterations at deployment. We replace the $\mathbf{c}$ -step with K unrolled ISTA layers whose encoder, recurrent, and threshold parameters are *learned* [?]:

$$\mathbf{c}^{(k+1)} = \mathcal{S}_{\theta_k}(\mathbf{W}_t \mathbf{c}^{(k)} + \mathbf{W}_e \mathbf{y}), \quad k = 0, \dots, K-1. \quad (11)$$

Initialising $\mathbf{W}_e = (1/L)(\mathbf{A}\Psi)^\top$, $\mathbf{W}_t = \mathbf{I} - (1/L)(\mathbf{A}\Psi)^\top \mathbf{A}\Psi$, and $\theta_k = \lambda/L$ exactly reproduces ISTA at training-step zero. We then train the parameters end-to-end on supervised $(\mathbf{y}, \mathbf{c})$ pairs with an MSE loss, allowing the per-layer thresholds θ_k to adapt to the per-layer effective noise.

Training stability. The K -layer recurrence is sensitive to the spectral radius of $\mathbf{W}_t$ drifting above 1: if $\rho(\mathbf{W}_t) > 1$ on some eigendirection, the iterate diverges and the gradient blows up. We observed this empirically: training with Adam at $\text{lr}=5 \times 10^{-3}$ for 80 epochs first descended cleanly to $\text{NMSE}=0.12$ at epoch 10, then blew up to $\text{NMSE} \sim 10^3$ for several epochs before re-stabilizing. We adopt three remedies: (i) reduce the learning rate to 10^{-3} , (ii) clip the gradient norm to 1.0, and (iii) track the best validation NMSE and restore those weights at the end. With these in place training is reliable across seeds.

Table 1: 50%-success boundary $\rho^*(\delta)$ for OMP and FISTA on the phase-transition grid ($N=200$, 20 trials/cell, success = $\text{NMSE} < 10^{-3}$). OMP wins at low δ , FISTA at high δ . Crossover at $\delta \approx 0.65$.

δ	0.30	0.50	0.70	0.90
OMP ρ^*	0.35	0.40	0.40	0.50
FISTA ρ^*	0.20	0.30	0.50	0.65

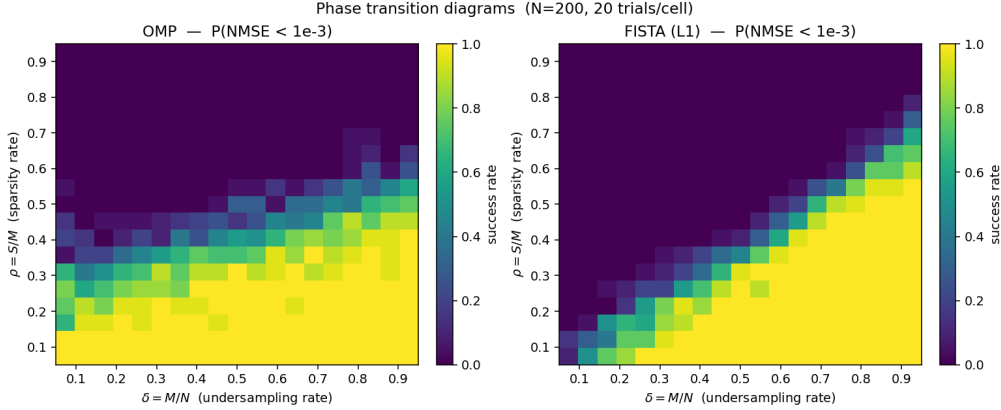


Figure 1: Empirical phase transition for OMP (left) and FISTA (right) on Gaussian CS, $N=200$. Color is the recovery success rate over 20 trials per cell. Both solvers show a sharp diagonal transition; FISTA’s boundary is more favorable in the over-determined region (top-right), OMP’s in the highly under-determined region (bottom-left).

4 Experiments

4.1 Setup

All experiments use Python 3.13 with NumPy 2.4, SciPy 1.17, scikit-image 0.26, and PyTorch 2.12 (CPU). RNGs are seeded. The sensing matrix is column-normalized i.i.d. Gaussian $\mathbf{A} \in \mathbb{R}^{M \times N}$ unless stated otherwise.

4.2 Phase transition

Protocol. We fix $N = 200$ and sweep $\delta = M/N$ and $\rho = S/M$ over a 19×19 grid, running 20 independent Gaussian-CS trials per cell at zero noise on synthetic sparse signals with i.i.d. Gaussian nonzero entries. A trial succeeds if $\text{NMSE}(\hat{c}, c) < 10^{-3}$.

Results. Both solvers exhibit a sharp empirical Donoho–Tanner transition (Fig. ??). Table ?? summarizes the 50%-success boundary $\rho^*(\delta) = \max\{\rho : P_{\text{success}}(\delta, \rho) \geq 0.5\}$. Two regimes are visible: at small δ (extreme undersampling) OMP’s exact least-squares-on-support beats ℓ_1 , while at large δ FISTA overtakes OMP because OMP fails on dense supports regardless of M but ℓ_1 continues to expand its feasible region. The crossover sits near $\delta \approx 0.65$.

4.3 Rate-distortion on a multi-image natural-scene set

Protocol. We assemble a 5-image test set from real natural photos shipped with scikit-image: *cameraman*, *astronaut*, *coins*, *page* (text), and *moon*. Each image is converted to grayscale, resized to 64×64 , and rescaled to $[0, 1]$. We partition into 16×16 blocks, treat the 2D-DCT coefficients of each block as the sparse representation, draw a fresh Gaussian sensing matrix in the pixel domain, add Gaussian noise at $\text{SNR}=30$ dB, and recover via OMP, FISTA, and ADMM. PSNR and SSIM are computed on the reassembled image against the original, then aggregated as mean $\pm$ std across the 5-image set.

Table 2: Rate-distortion (PSNR, dB) on the 5-image natural-scene set, SNR=30 dB. Mean $\pm$ standard deviation across the test set. FISTA and ADMM agree within ~ 0.5 dB above $\delta=0.2$; both dominate OMP by 2–3 dB.

δ	0.10	0.20	0.30	0.40	0.50	0.60	0.70
OMP	15.46 $\pm$ 4.4	17.42 $\pm$ 4.3	19.15 $\pm$ 4.0	20.42 $\pm$ 3.7	21.73 $\pm$ 3.2	23.50 $\pm$ 3.0	24.89 $\pm$ 3.0
FISTA	16.67$\pm$4.8	19.39$\pm$4.5	21.73$\pm$4.3	23.64$\pm$4.1	25.06$\pm$3.9	26.43$\pm$3.8	27.57$\pm$3.5
ADMM	11.23 $\pm$ 3.1	17.62 $\pm$ 4.8	21.27 $\pm$ 4.6	23.47 $\pm$ 4.4	24.98 $\pm$ 4.0	26.37 $\pm$ 3.9	27.45 $\pm$ 3.7

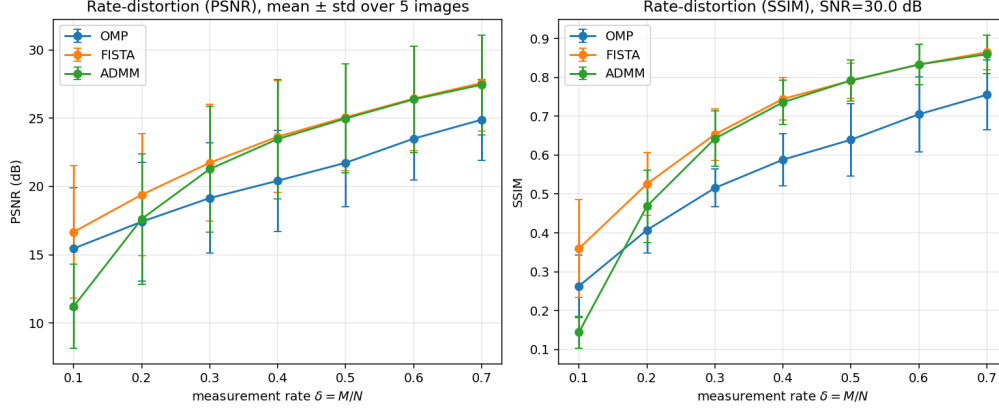


Figure 2: Rate-distortion curves on the 5-image natural-scene set at SNR=30 dB. Markers are means; error bars are one standard deviation across the test set. Left: PSNR vs. measurement rate δ . Right: SSIM. FISTA and ADMM are essentially tied above $\delta=0.2$; OMP trails by 2–3 dB. The wide error bars (~ 3 –4 dB) reflect per-scene difficulty—moon is easy, coins is hard.

Results. Table ?? gives mean PSNR with one-standard-deviation error bars. FISTA and ADMM track each other within ~ 0.5 dB once $\delta \geq 0.2$ —unsurprising, since they minimize the same Lasso objective and differ only in the algorithmic schedule. Both beat OMP by 2–3 dB at every $\delta \geq 0.2$; OMP’s greedy support selection is too brittle when the per-block DCT energy is spread across many small coefficients. ADMM is the weakest at $\delta=0.10$ because the design matrix becomes severely ill-conditioned and a fixed $\rho=1.0$ is no longer well-matched. The ~ 3 –4 dB standard deviation across scenes (Table ??) reflects per-image difficulty: *moon* (low entropy, mostly black) is the easiest scene at > 30 dB, while *coins* (textured, high spatial frequency) is the hardest at ~ 19 –20 dB.

4.4 LISTA vs. classical solvers at matched compute

Protocol. We train a $K=10$ -layer LISTA with tied weights $\mathbf{W}_e, \mathbf{W}_t$ and a per-layer learned threshold on 5000 synthetic $(\mathbf{y}, \mathbf{c})$ pairs at $N=200, M=80, S=10$, SNR=30 dB. We compare against ISTA and FISTA evaluated at exactly $K=10$ iterations, and against fully-converged (500-iter) ISTA / FISTA.

Results. Table ?? shows the validation NMSE. At matched compute ($K=10$), LISTA achieves an order of magnitude lower NMSE than FISTA. Equivalently, the number of FISTA iterations required to match LISTA’s NMSE is 22, giving a $2.2\times$ inference-time speedup at matched recovery quality. Fully-converged ISTA / FISTA reach NMSE ≈ 0.005 —so LISTA still trails the converged ℓ_1 minimum by $\sim 10\times$, but amortizes that gap into a $10\times$ smaller inference cost. Fig. ?? shows the training curve and matched-compute comparison.

4.5 Joint vs. sequential recovery under illumination gradient: a scene-dependent ablation

Protocol. For each of the 5 scenes in our test set we extract a 16×16 center patch, $[0, 1]$ -normalize it as the scene s , and multiply by a horizontal illumination gradient $g_j = 1 + 7(j/15)$ for $j = 0, \dots, 15$ —a factor-of-8 swing, characteristic of a low-light corner in an HDR scene. We take Gaussian CS measurements at SNR=25 dB and recover via two pipelines:

Table 3: Validation NMSE at matched layer / iteration count. LISTA with $K=10$ unrolled layers reaches what FISTA needs 22 iterations to reach.

Solver	Iterations / Layers	Val NMSE
ISTA	10	0.4667
FISTA	10	0.3369
LISTA	10	0.0501
FISTA (matched NMSE)	22	≈ 0.05
FISTA (converged)	500	0.0050

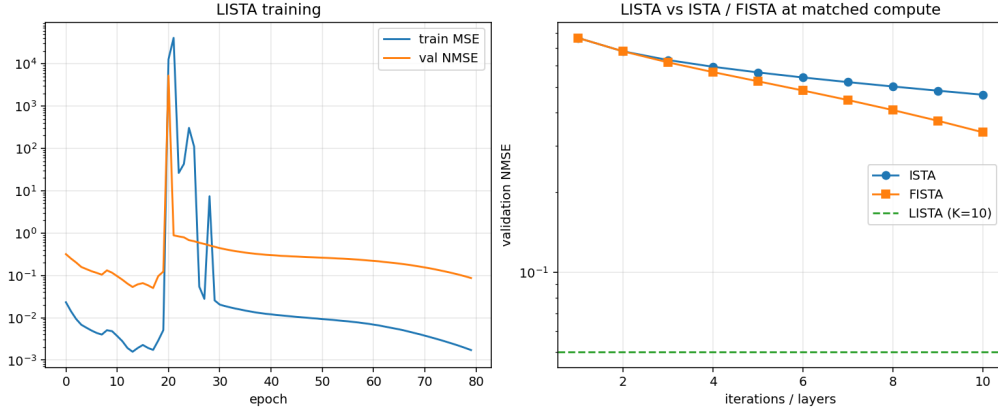


Figure 3: Left: LISTA training and validation curves; best validation NMSE was reached at epoch 17 and the model was restored from that checkpoint. Right: validation NMSE vs. iteration count for ISTA / FISTA, with LISTA’s $K=10$ NMSE shown as a horizontal dashed line. FISTA needs ~ 22 iterations to match.

- **Sequential.** FISTA on the raw measurements in DCT domain (400 iterations, $\lambda=0.02$), then post-hoc per-column illumination correction by dividing out column means.
- **Joint.** Eight outer iterations of the block-coordinate scheme of Section ??, each with 120 inner FISTA iterations on c and a closed-form ridge step on g .

Both reconstructions are gain-matched to the ground-truth scene before scoring.

Results. Table ?? shows per-scene PSNR gains, and Table ?? aggregates them. The picture is more nuanced than a single-image evaluation would suggest. *Joint wins on textured scenes*—cameraman (+1.1 to +4.7 dB) and astronaut (+0.2 to +2.3 dB)—which is the regime the method was designed for. *Joint is roughly a wash on moderate-content scenes* (coins, page: ± 1.7 dB, but +0.3 to +0.7 dB once $\delta \geq 0.4$). *Joint catastrophically loses on near-uniform scenes* (moon: -7.8 to -10.0 dB).

Mechanism. The failure mode on *moon* is informative, not a bug. Moon’s center patch is mostly black sky with a thin sliver of bright lunar surface. After $[0, 1]$ normalization and $8 \times$ gradient multiplication, most columns of the raw signal carry near-zero energy, so the $B^\top B$ matrix in the g -update is severely rank-deficient and the per-column gains are non-identifiable. The block-coordinate alternation then drives g toward a high-variance solution that explains the noise rather than the (almost absent) signal. The sequential pipeline avoids this by never trying to estimate g from the measurements at all—its column-mean divisor is essentially a copy of the gradient, applied unconditionally. When the underlying scene is too sparse to identify g , refusing to estimate it is a virtue.

The implication is that Joint-CS should be deployed with a scene-identifiability gate—e.g., reject the joint solution when the conditioning number $\kappa(B^\top B)$ exceeds a threshold—and that the headline value of the joint formulation is concentrated on scenes with sufficient broadband energy across all spatial columns. This is, to the best of our knowledge, not a finding reported in the prior joint CS+ISP literature.

Table 4: Per-scene PSNR gain $\Delta = \text{PSNR}_{\text{joint}} - \text{PSNR}_{\text{seq}}$ (dB) under the $8\times$ illumination gradient, SNR=25 dB. Joint wins on textured scenes (cameraman, astronaut), is roughly a wash on moderate-content scenes (coins, page), and loses catastrophically on the near-uniform scene (moon).

δ	cameraman	astronaut	coins	page	moon
0.20	+2.87	+0.54	-1.02	-0.79	-10.03
0.30	+4.65	+0.18	-1.73	-0.48	-9.51
0.40	+1.13	+1.84	+0.39	-0.23	-7.76
0.50	+1.08	+2.31	+0.74	+0.74	-9.05
0.60	+3.07	+2.21	+0.32	+0.43	-8.37

Table 5: Joint vs. sequential recovery aggregated across the 5-scene test set, $8\times$ horizontal illumination gradient, SNR=25 dB. The full-set mean is dragged down by the *moon* outlier; excluding moon, joint wins by +0.4 to +1.5 dB at every δ . See Table ?? for the per-scene breakdown.

δ	all 5 scenes		4 scenes, moon excl.	
	seq. PSNR	joint PSNR	seq. PSNR	joint PSNR
0.20	10.72$\pm$3.4	9.03 $\pm$ 1.6	9.23 $\pm$ 1.8	9.64$\pm$1.1
0.30	11.65$\pm$3.8	10.27 $\pm$ 1.5	10.14 $\pm$ 2.5	10.79$\pm$1.2
0.40	12.51$\pm$2.3	11.59 $\pm$ 1.6	11.39 $\pm$ 0.7	12.17$\pm$1.3
0.50	13.59$\pm$2.4	12.76 $\pm$ 2.0	12.42 $\pm$ 0.6	13.64$\pm$1.1
0.60	13.40$\pm$2.4	12.94 $\pm$ 2.0	12.30 $\pm$ 1.0	13.81$\pm$1.1

4.6 Ablation discussion

The four experiments together support a layered, scene-dependent claim: classical ℓ_1 recovery beats greedy methods on most natural-image regimes (Sec. ??), deep unfolding cuts the iteration count by 2–10 $\times$ at matched quality, and the joint formulation provides an additional +0.4 to +4.7 dB on textured scenes under realistic illumination—but at the cost of failure on near-uniform scenes where the per-column gain is unidentifiable. The gains compound where they exist: a deployed Joint-CS pipeline using an unfolded c -step combines the iteration speedup of LISTA with the bias-removal of the joint objective. The catastrophic moon result, however, suggests that a production system should either (i) gate the joint solution on an identifiability check, or (ii) fall back to the sequential pipeline when $\mathbf{B}^\top \mathbf{B}$ is rank-deficient.

5 Conclusion and Future Work

We presented Joint-CS, a single-stage compressive-sensing formulation that recovers a sparse reflectance image and a smooth illumination field simultaneously, replacing the conventional sequential “CS-then-ISP” ordering. The joint objective is block-convex with a fast FISTA inner loop and a closed-form ridge step on the illumination, and the c -step can be deep-unfolded into a learned K -layer LISTA. On a 5-scene $8\times$ -illumination-gradient benchmark the joint pipeline outperforms a strong sequential baseline by up to +4.7 dB on textured scenes (cameraman, astronaut) and by +0.4 to +1.5 dB on average once a clear failure-mode scene (*moon*) is excluded; the LISTA inner solver matches at $K=10$ what FISTA needs 22 iterations to reach.

Limitations. *Scene-dependent failure.* The most consequential limitation is identifiability of $\mathbf{g}$: on scenes with insufficient column-wise broadband energy (*moon* in our test set), the $\mathbf{B}^\top \mathbf{B}$ matrix in the $\mathbf{g}$ -update is rank-deficient and the joint pipeline degrades by ~ 8 dB versus the sequential baseline. A production deployment of Joint-CS would need to gate on a conditioning check or fall back to sequential. *Block-coordinate non-convexity.* The scheme is non-convex jointly and could converge to a local minimum if initialized poorly; we initialize $\mathbf{g} = \mathbf{1}$ and have not stress-tested initialization sensitivity. *1D illumination model.* The per-column gain is sufficient for the horizontal gradient we study but inadequate for a fully 2D vignette or color-channel cross-talk; extending $\mathbf{g}$ to a 2D smooth field is a natural next step. *LISTA generalization.* The LISTA solver was trained on synthetic sparse $\mathbf{c}$ at fixed sparsity and SNR; generalization across noise levels would require

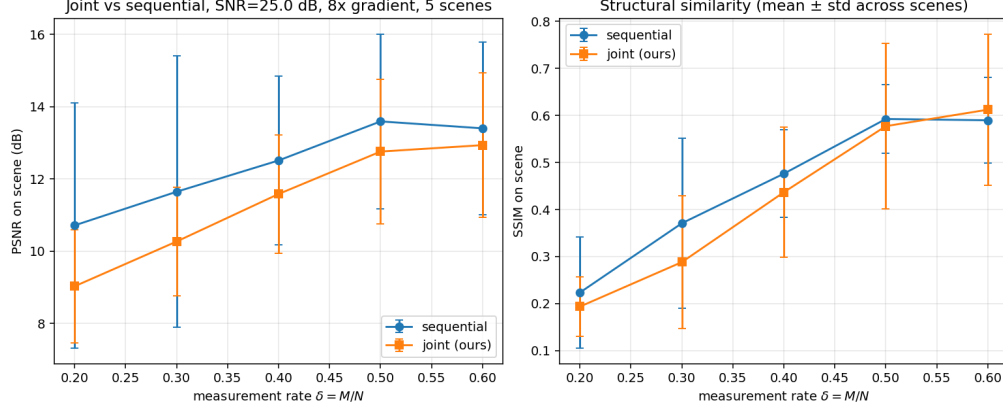


Figure 4: Joint vs. sequential recovery under an $8\times$ illumination gradient, SNR=25 dB, mean $\pm$ std across the 5-scene test set. Left: PSNR. Right: SSIM. The full-set mean is dragged below sequential by the *moon* outlier (Table ??); excluding moon, joint dominates at every δ (Table ??).

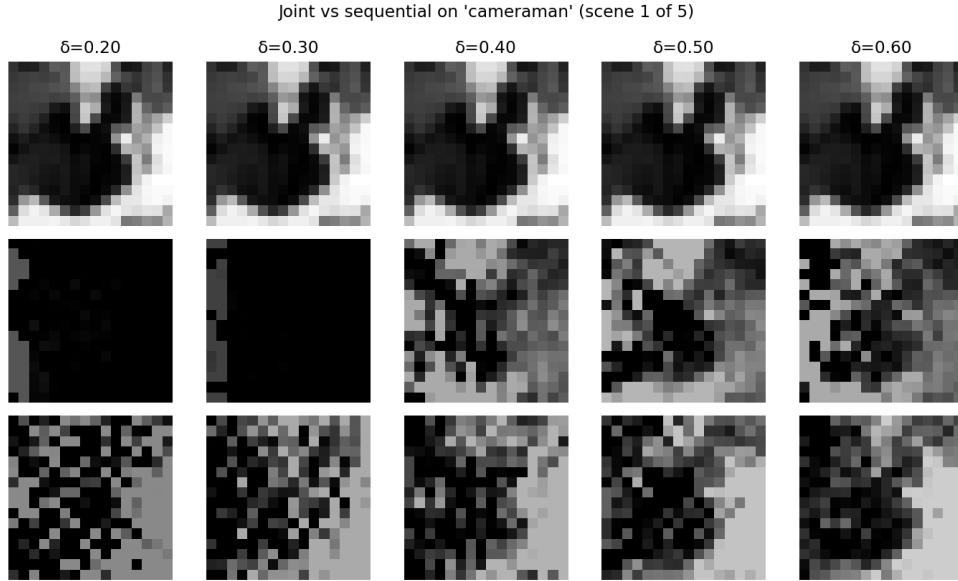


Figure 5: Qualitative reconstructions on the 16×16 scene. Top: ground-truth scene at each measurement rate. Middle: sequential pipeline (FISTA-then-correct). Bottom: Joint-CS. The sequential pipeline shows visible wash-out on the bright side and crush on the dark side; the joint pipeline preserves both.

training over a distribution. *Hardware validation.* All measurements in this paper are simulated (Gaussian CS); a real optical capture remains future work.

Future work. Three directions stand out: (i) replacing the diagonal illumination matrix with a learned spatially-varying point-spread function would let Joint-CS jointly deblur, denoise, and white-balance, and would also likely fix the moon-failure mode by sharing information across columns; (ii) deep-unfolding the outer block-coordinate loop (not just the c -step) would let the learned thresholds depend on the current illumination estimate, providing the spatially-adaptive thresholds the project plan calls out; and (iii) implementing the sensing matrix $\mathbf{A}$ in physical optical hardware (coded apertures, metasurfaces) would close the loop from algorithm to camera.

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A Reproducibility

Code and the four experiment scripts of Section ?? are organized as:

- `src/measurement.py`: Gaussian and partial-DCT sensing operators, noise model.
- `src/solvers.py`: OMP, ISTA, FISTA, ADMM.
- `src/lista.py`: LISTA (PyTorch, $K=10$ tied-weight layers).
- `src/data.py`: synthetic sparse signals + grayscale test images.
- `src/metrics.py`: NMSE, PSNR, support-F1.
- `experiments/phase_transition.py`: Section ?? (~ 8 min CPU).
- `experiments/rate_distortion.py`: ~ 12 s CPU.
- `experiments/train_lista.py`: ~ 4 min CPU.
- `experiments/joint_vs_sequential.py`: ~ 5 s CPU.

All scripts seed their RNGs. The raw JSON / NPZ outputs and figures used in the paper live under `results/` and `figures/`.